**2021-Ph-H1-Q12**

**Section: Particles and Waves**

**Topic: Forces on Charged Particles**

A charged particle moves in a magnetic field. What happens?

**Solution:**

Charged particles in a magnetic field experience a **centripetal force** causing **circular motion** (if entering perpendicular to field).

**Answer: C**

**Guidance for Students:**

The magnetic force acts **perpendicular** to the velocity, causing circular motion if the field is uniform and the entry is at right angles.

**Revision Tip:**

Remember:

$F = qvB \quad \text{and the force is always} \;\; \text{to velocity.}$